Flat-Rate Medicaid Reimbursement and Geographic Variation in Provider Participation: Evidence from California

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Data Availability: County- and specialty-level data are publicly available at https://caphegroup.org/tools/access-explorer

Abstract

Objective: To quantify geographic and specialty-level variation in Medi-Cal provider participation rates across California's 58 counties and examine the relationship between local practice costs and provider engagement with the program.

Data Sources: NPPES provider registry (2024); HHS Medicaid Provider Spending dataset (January 2018–December 2024); composite practice cost index from BLS wage data, HUD Fair Market Rents, and healthcare purchased services indices.

Study Design: Descriptive analysis of provider participation rates, pooled over a seven-year period (2018–2024), defined as the share of NPPES-registered providers in each county who actively billed Medi-Cal during the study period. Rates were stratified by county, Hospital Referral Region (HRR), and six specialty categories.

Principal Findings: Among 79,223 NPPES-registered providers across California's 58 counties, 31,798 (40.1%) actively billed Medi-Cal. County-level participation ranged from 0% (Sierra County, N=2 providers) to 48.3% (Marin County). Bay Area counties with the highest practice costs (composite cost index 162–186) paradoxically exhibited higher participation (44–48%) than low-cost rural counties (17–30%), suggesting that provider density and institutional infrastructure may matter more than reimbursement adequacy alone. Behavioral health had the lowest median participation (27.3%) across all specialties, with the vast majority of counties clustered between 25% and 29%. Low overall participation (<34%) clusters along the Central Valley and far-north I-5 corridor, with four HRRs (Bakersfield, Modesto, Stockton, and Redding) covering approximately 3.6 million residents.

Conclusions: Medi-Cal's flat-rate reimbursement, which unlike Medicare's GPCI makes no geographic adjustment, produces predictable gaps in provider participation. The positive association between practice costs and participation suggests urban institutional infrastructure buffers against low reimbursement in ways unavailable to rural communities. Geographic payment adjustment and targeted behavioral health investments warrant consideration as states implement new CMS network adequacy requirements. Data are available at https://caphegroup.org/

tools/access-explorer.

1 Introduction

Medicare adjusts physician payments by geography through the Geographic Practice Cost Index (GPCI), which accounts for local variation in practice costs across three components: physician work, practice expense, and malpractice insurance.[1] California's Medicaid program, Medi-Cal, applies no such adjustment to its physician fee schedule. A primary care visit in San Francisco, where office rent may exceed three times the statewide average, reimburses at the same rate as one in Modoc County, where practice costs are roughly 76% of the state mean.

This flat-rate structure has long raised questions about whether it creates geographic disparities in provider willingness to participate in the program. Provider participation, defined here as the share of licensed providers who actively bill Medicaid, is a broader measure than the "phantom network" concept examined by Zhu and colleagues, who documented that 58% of behavioral health providers listed in Oregon Medicaid directories were "phantom" providers with no Medicaid claims.[2] Where phantom network studies focus on the discrepancy between listed and available providers within a plan's directory, participation rates capture the upstream question: what fraction of the total provider workforce engages with Medicaid at all?

Recent federal action has heightened urgency. In May 2024, the Centers for Medicare and Medicaid Services (CMS) finalized CMS-2439-F, a rule requiring states to verify managed care provider directories against actual utilization data, moving beyond paper compliance toward demonstrated network adequacy.[3] The HHS Office of Inspector General subsequently reported that 45% of surveyed behavioral health providers were unavailable to treat new Medicare and Medicaid enrollees, with three-quarters citing full caseloads.[4] And new data on Medicaid-to-Medicare fee ratios show a persistent national gap: Medicaid physician fees remained at approximately 71% of Medicare rates in 2024, with minimal improvement over two decades.[5]

Despite this policy attention, no publicly available tool quantifies county-by-county, specialty-by-specialty Medicaid participation rates in California. This analysis fills that gap, drawing on provider registry and billing data to map participation across all 58 counties and six specialty categories, and examining whether the relationship between local practice costs and provider engagement follows the pattern that flat-rate reimbursement would predict.

2 Methods

2.1 Data Sources

Provider registration data came from the NPPES, which maintains the national registry of all providers with active National Provider Identifiers (NPIs).[6] We identified all providers with active California practice addresses as of the December 2024 monthly file, classified by Healthcare Provider Taxonomy codes.

Active Medi-Cal participation was determined using the HHS Medicaid Provider Spending dataset, which aggregates individual claims to the provider-procedure-month level for all fee-for-service and managed care Medicaid billing.[7] A provider was classified as an active Medi-Cal participant if they submitted at least one paid claim between January 2018 and December 2024.

2.2 Participation Rate Definition

The participation rate for each county was calculated as:

Participation Rate = Active Medi-Cal Billers / Total NPPES-Registered Providers $\times$ *100*

This measure captures the extensive margin of participation (whether a provider engages with the program at all) rather than the intensive margin of how many Medicaid patients a participating provider serves.

2.3 Specialty Classification

Providers were assigned to six categories based on Healthcare Provider Taxonomy codes: Primary Care (family medicine, internal medicine, pediatrics, general practice), Behavioral Health (psychiatry, psychology, clinical social work, licensed counseling), Dental (general dentistry, oral surgery, orthodontics, pediatric dentistry), OB/GYN (obstetrics, gynecology, maternal-fetal medicine), Other Surgical (all surgical subspecialties not otherwise classified), and Pharmacy and Durable Medical Equipment (DME). Providers with multiple taxonomy codes were assigned to their primary classification.

2.4 Composite Practice Cost Index

To examine the relationship between local practice costs and provider participation, we constructed a composite practice cost index for each county. The index incorporates three major practice expense categories drawn from the Medicare Economic Index framework: healthcare employee wages (from Bureau of Labor Statistics Quarterly Census of Employment and Wages), facility rent (from HUD Fair Market Rents), and purchased services (from regional healthcare services price indices), weighted to reflect the relative contribution of each category to overall practice costs. Each component is benchmarked to the statewide average (index = 100). An effective reimbursement index was derived as the inverse: 100 / composite cost index $\times$ 100, representing the relative purchasing power of a flat Medi-Cal payment in each county.

2.5 Geographic Units

County-level analysis covered all 58 California counties. Hospital Referral Region (HRR) analysis used 24 HRRs as defined by the Dartmouth Atlas, with county-to-HRR mapping based on plurality assignment of ZIP codes.[8]

2.6 Limitations

Several limitations should be noted. First, participation as defined here differs from phantom network measures, which use plan directory listings rather than the full NPPES registry as the denominator. Our measure is intentionally broader. Second, although the billing data span seven years (2018–2024), participation rates are pooled across the full period rather than computed annually;

the analysis therefore cannot track trends in provider engagement over time. Third, billing at least one claim over a seven-year window sets a low threshold for "active" participation; some providers classified as active may have had only minimal Medicaid engagement.

3 Results

3.1 Statewide Overview

Among 79,223 NPPES-registered providers with active California practice addresses, 31,798 (40.1%) actively billed Medi-Cal between January 2018 and December 2024. The statewide weighted participation rate masks substantial variation across geographies and specialties.

3.2 Geographic Variation

County-level participation rates ranged from 0% in Sierra County (N=2 registered providers, zero active) to 48.3% in Marin County (Exhibit 1). The ten highest-participation counties were exclusively in the Bay Area and coastal Southern California, with rates from 40.5% (Sonoma) to 48.3% (Marin). All ten had composite cost indices above 145, meaning local practice costs exceeded the statewide average by at least 45%.

The ten lowest-participation counties were entirely rural and frontier, with rates from 0% (Sierra) to 25.9% (Tehama). These counties had cost indices below 91, meaning Medi-Cal's flat reimbursement actually stretches further in purchasing power. Yet few providers engage with the program, suggesting that geographic isolation, small provider pools, and limited institutional infrastructure drive non-participation in rural areas more than reimbursement adequacy alone (Exhibit 3).

At the HRR level, participation ranged from 30.0% in the Redding HRR (covering seven rural northern counties) to 47.0% in the San Francisco HRR. Three tiers emerged: high participation (>40%) across eight Bay Area and coastal HRRs; moderate participation (34–40%) across eleven

Central Valley, Inland Empire, and metropolitan HRRs including Los Angeles (38.5%); and low participation (<34%) across five Central Valley and far-north HRRs. Provider density disparities compound these patterns: the San Francisco HRR has 4.51 registered providers per 1,000 residents compared with 1.15 per 1,000 in the Bakersfield HRR, a nearly 4:1 ratio.

3.3 Specialty Variation

Participation varied substantially by specialty (Exhibit 4). Pharmacy and DME providers had the highest median participation (72.2%), likely reflecting different business models in which Medicaid billing is routine. Primary care participation clustered around the median of 41.4%, with relatively modest county-to-county variation. Behavioral health had the lowest median participation at 27.3%, meaning nearly three in four registered behavioral health providers in the typical county did not bill Medi-Cal, consistent with national findings that psychiatrist Medicaid acceptance rates have declined to approximately 36%.[9]

The behavioral health crisis is notable for its uniformity rather than extreme outliers: 44 of 57 reporting counties clustered between 25% and 30% participation. Only two counties fell below 20%: Alpine (0%, N=1 registered) and Imperial (12.3%, N=65 registered). Imperial County stands out because it has a provider pool large enough for the rate to be meaningful; its 12.3% behavioral health participation suggests systemic barriers, potentially related to its border-region location and demographic composition. The contrast with pharmacy participation (statewide median 72.2%) underscores that specialty-specific factors, not general county characteristics, drive behavioral health non-participation (Exhibit 2).

3.4 Cost-Participation Relationship

The relationship between practice costs and Medi-Cal participation runs counter to the prediction of a simple reimbursement-adequacy model. Under such a model, counties where flat-rate payments buy less (high cost index, low effective reimbursement index) should exhibit lower participation. Instead, the data show the opposite: the correlation between composite cost index and participation rate is positive. Marin County (cost index 172.8, effective reimbursement index 57.9) leads in participation at 48.3%, while Modoc County (cost index 75.9, effective reimbursement index 131.8) has the lowest non-zero rate at 16.7%.

This pattern suggests that urban institutional infrastructure (academic medical centers, FQHCs, large multispecialty groups with diversified payer mixes) provides a buffer that sustains Medicaid participation despite low reimbursement. FQHCs receive prospective Medicaid reimbursement under a cost-related payment system independent of the physician fee schedule, and academic medical centers cross-subsidize Medicaid care with commercial and research revenue. Rural areas lack these institutional buffers, leaving independent practitioners to absorb the full impact of below-cost reimbursement (Exhibit 3).

3.5 Critical Access Alerts

Five counties had at least one specialty with a participation rate below 20%: Alpine (primary care, behavioral health, and dental at 0%), Imperial (behavioral health at 12.3%), Mariposa and Modoc (OB/GYN at 0%), and Sierra (primary care at 0%). Small-population frontier counties account for most of these alerts, where a single provider entering or leaving the workforce can shift rates dramatically. The exception is Imperial County, whose 268 registered providers make its 22.4% overall participation rate and 12.3% behavioral health rate a more reliable indicator of systemic access challenges.

4 Discussion

Medi-Cal's flat-rate reimbursement structure produces stark geographic and specialty-level disparities in provider participation, though the mechanism differs from what a simple reimbursement-adequacy model would predict.

The counterintuitive positive correlation between practice costs and participation points to a structural explanation: high-cost urban counties concentrate the institutional infrastructure (academic medical centers, FQHCs, and large health systems) that sustains Medicaid participation through cross-subsidization and mission-driven care. FQHCs, which receive prospective Medicaid reimbursement under a cost-related payment system independent of the physician fee schedule, serve both urban and rural communities. Academic medical centers cross-subsidize Medicaid care with commercial and research revenue, and large multispecialty groups can absorb low-reimbursement patients within diversified payer mixes. Rural counties, lacking these institutional buffers, leave independent practitioners to absorb the full impact of below-cost reimbursement despite nominally higher purchasing power from flat-rate payments. MACPAC research supports this interpretation: physician practices with higher Medicaid caseloads were more willing to accept new Medicaid patients than practices with below-average caseloads, suggesting that organizational characteristics and existing Medicaid patient volume predict acceptance more strongly than payment rates alone.[10]

The pattern also contextualizes California's recent commitments under its Section 1115 demonstration waiver, which requires annual increases of 2 percentage points in the Medicaid-to-Medicare rate ratio toward a threshold of 80% of Medicare rates for primary care, obstetric, and behavioral health services.[5] The evidence from this analysis suggests that reaching the 80% threshold may improve access in urban areas where institutional infrastructure can channel payment increases into expanded capacity, but may prove insufficient in rural counties where the binding constraint is provider supply rather than payment levels.

The behavioral health findings are especially concerning. With median participation at 27.3% and 44 of 57 reporting counties clustered between 25% and 30%, the crisis is statewide rather than concentrated in a few outlier counties. This uniformly low participation mirrors national trends documented by Wen and colleagues, who found psychiatrist Medicaid acceptance declining from 47.9% to 35.4% between 2010 and 2015 despite concurrent Medicaid expansion.[9] Zhu and colleagues have shown that Medicaid reimburses psychiatric services at a median of 76% of Medicare rates, with substantial state-to-state variation that does not correlate well with the supply of Medicaid-participating psychiatrists, suggesting that payment rates alone may be insufficient to overcome other barriers to participation at current levels.[11] The HRSA Behavioral Health Workforce Brief projects severe national shortages across psychologists (48% adequacy by 2038), mental health counselors (55% adequacy), and addiction counselors (30% adequacy), compounding the payment-driven barriers documented here.[12] The narrow range of behavioral health participation across counties of vastly different sizes and cost structures suggests a systemic ceiling on Medi-Cal behavioral health engagement that persists regardless of local conditions.

4.1 Policy Implications

Three policy directions emerge from these findings.

First, a geographic payment adjustment for Medi-Cal, analogous to Medicare's GPCI, warrants consideration, though the cost-participation relationship observed here suggests such an adjustment alone would be insufficient in rural areas. Research on the ACA's temporary primary care fee increase found mixed results: while the majority of studies found some association between the payment increase and improved access, effects were concentrated among existing participants rather than newly recruited providers.[10] Rural-specific interventions such as enhanced FQHC placement, telehealth infrastructure, loan forgiveness programs, and training pipeline development may be more effective than across-the-board payment increases in the lowest-cost counties where provider supply, not payment adequacy, represents the binding constraint.

Second, the new CMS network adequacy verification requirements under CMS-2439-F should be implemented with specialty-specific standards. Aggregate adequacy measures would mask the severe behavioral health deficits identified here; a county meeting overall network standards could still have critical behavioral health gaps. With median behavioral health participation at just 27.3% statewide, even counties that appear adequate in aggregate are likely failing reasonable specialty-specific standards when behavioral health is measured separately from pharmacy and primary care.

Third, the interactive Access Explorer tool (https://caphegroup.org/tools/access-explorer that underlies this analysis enables ongoing, public monitoring of provider participation patterns. Making participation data routinely available at the county and specialty level could support evidence-based workforce planning and accountability under the new CMS requirements.

4.2 Limitations

Beyond the methodological limitations noted above, this analysis cannot distinguish between providers who choose not to participate in Medi-Cal and those who are functionally unavailable due to retirement, administrative status, or practice outside clinical care. The NPPES registry includes all active NPIs regardless of clinical activity. Additionally, participation rates cannot directly measure patient access; a county with a 40% participation rate may have adequate access if participating providers maintain open panels, while a county with a 30% rate may face severe shortages if those providers are at capacity. The HHS OIG finding that three-quarters of unavailable behavioral health providers cited full caseloads suggests that participation rates likely overstate functional access.[4]

5 Conclusion

Medi-Cal's uniform reimbursement structure generates predictable geographic variation in provider participation, with rural and frontier counties experiencing the lowest engagement despite nominally favorable reimbursement-to-cost ratios. The positive association between practice costs and participation suggests that institutional infrastructure, more than payment levels alone, sustains the Medicaid safety net in urban areas. Behavioral health participation represents a statewide crisis, with three-quarters of registered providers not billing Medi-Cal in the typical county. As CMS implements new network adequacy verification requirements, granular, publicly available participation data of the kind presented here could inform both state compliance strategies and evidence-based workforce policy.

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Exhibit 1. Medi-Cal Provider Participation Rate by County, California, 2018–2024

Map of California's 58 counties shaded by overall Medi-Cal provider participation rate. Darker shading indicates higher participation. Bay Area and coastal Southern California counties show the highest rates (40–48%), while rural northern and Central Valley counties show the lowest (<30%).

[Figure available in the online version of this paper.]

Exhibit 2. Behavioral Health vs. Pharmacy/DME Provider Participation by County, California, 2018–2024

Side-by-side bar chart comparing behavioral health (median 27.3%) and pharmacy/DME (median 72.1%) participation rates across the 20 most populous counties. Pharmacy/DME participation (median 72.2%) far exceeds behavioral health (median 27.3%) across all counties, while behavioral health falls below 30% in most.

[Figure available in the online version of this paper.]

Exhibit 3. Top 10 and Bottom 10 Counties by Medi-Cal Provider Participation Rate

Rank	County	Rate (%)	Registered	Active	Cost Index	Eff. Reimb. Index
1	Marin	48.3	620	299	172.8	57.9
2	San Francisco	46.8	4,210	1,970	186.0	53.8
3	San Mateo	45.1	1,820	821	181.1	55.2
4	Santa Clara	44.9	5,640	2,532	176.7	56.6
5	Alameda	44.2	5,830	2,577	162.1	61.7
6	Contra Costa	43.8	3,210	1,406	157.6	63.5
7	Orange	42.7	8,420	3,595	154.8	64.6
8	Napa	41.2	245	101	145.1	68.9
9	San Diego	41.2	8,903	3,668	149.6	66.8
10	Sonoma	40.5	685	277	152.8	65.4
49	Tehama	25.9	58	15	85.0	117.6
50	Mariposa	25.0	12	3	90.8	110.1
51	Colusa	25.0	16	4	85.6	116.8
52	Lake	24.1	58	14	89.8	111.4
53	Plumas	23.1	13	3	87.9	113.8
54	Imperial	22.4	268	60	83.7	119.5
55	Lassen	22.2	27	6	82.1	121.8
56	Trinity	20.0	15	3	79.6	125.6
57	Modoc	16.7	12	2	75.9	131.8
58	Sierra	0.0	2	0	79.9	125.2

Note: Composite cost index benchmarked to statewide average = 100. Effective reimbursement index = 100/composite cost index $\times$ 100, representing relative purchasing power of flat Medi-Cal payment. Data from NPPES (December 2024) and HHS Medicaid Provider Spending dataset (January 2018–December 2024).

Exhibit 4. Medi-Cal Provider Participation by Specialty Category, Statewide, 2018–2024

Specialty	Median Rate (%)	Registered	Active	Counties
Pharmacy/DME	72.2	6,678	4,819	57
Primary Care	41.4	20,372	8,449	58
OB/GYN	38.4	9,308	3,572	56
Dental	35.9	14,009	5,017	57
Other Surgical	30.8	12,136	3,731	56
Behavioral Health	27.3	16,720	4,563	57

Note: Median computed across reporting counties. Not all counties have providers in all specialty categories; Alpine and Sierra counties are missing from several categories due to very small provider populations. Data from NPPES (December 2024) and HHS Medicaid Provider Spending dataset (January 2018–December 2024).